2025 Ligabue: Airy's Experiment and Relativity — verification companion

Franco Ligabue

Ligabue 2025 — Airy's Experiment and Relativity (verification companion)

> [!abstract] Legend - 💡 tip/note — definitions, how a value is used - ⚠ warning — where the sheet's old labels disagreed with the slides - 📌 quote — verbatim from the slides / vault breakdown

> [!success] PDF present `project_sources/2025_Ligabue_Airy_Relativity.pdf` (14 slides) is in this folder — check the slide/equation references below directly. Full slide-by-slide transcription is in the vault: `/home/alan/Documents/multi_2/Sources/2025_Ligabue_A` (slide images `slide-01.png` … `slide-14.png`).

Bibliographic Info

- **Author:** Franco Ligabue
- **Title:** Airy's Experiment and Relativity
- **Year:** 2025
- **Form:** slide deck, 14 slides (`airy_relativity-1.pdf`)

Formulas used in the sheet / notebook, pinned to the slides

Every number in the AIR and WATER 4-vector blocks traces to one of these. Symbols: $\beta = v/c \approx 9.94\times10^{-5}$ ($v \approx 29.79$ km/s), $n = 1.33$ (water), angles in arcsec (× 206264.806).

Slide	Eq	Statement (verbatim / near-verbatim)	Where it is used
1	—	$k = 2\pi/\lambda = n\omega/c$; phase = group velocity = c/n in a non-dispersive medium	sets |k|=1 (air) and |k|=n=1.33 (water): ky = −1 vs −n
2	—	$k^\mu \equiv (\omega/c,\ k)$; phase $k\cdot r - \omega t$ is a Lorentz scalar	the 4-vector rows kt,kx,ky,kz
2	(3)	$k'_x = \gamma(k_x + \beta\omega/c)$	the boost that turns kx=0 → kx=∓β (AIR moving columns)
2	(4)	$k'_y = k_y$	ky unchanged by the x-boost
2	(5)	$\omega' = \gamma(\omega + \beta c\cdot k_x)$	kt row

Slide	Eq	Statement (verbatim / near-verbatim)	Where it is used
2	(6)	`tanθ' = γ(sinθ + β/n)/cosθ`	general angle transform behind every "ray bending" cell
3	(7)	`sinθ = −β/n`; for n=1, `sinθ = −β` (standard aberration)	AIR: rest frame 0″, moving frame arctan(β)=20.5″. WATER Sun rest: arcsin(β/n) = **15″**
13	—	Summary slide. Boundary conditions "can be done easily in the water's rest frame"; Snell "in the frame where water is stationary"; **" 'Timing' reasonings are incorrect"**; any telescope-angle argument ignoring boundary conditions is "flawed"	the rest-frame retreat (matches Pauli p.114, Rosser p.154)
14	—	Sun-frame water aberration: `θ₂ = −β/n ≈ −15″` (medium's frame), `θ₂' = θ₂ + βn = β(n − 1/n) ≈ 12″` (Sun's frame); plane waves give 0; **"the value of the speed… will not be c/1.33"**	WATER: ⊙ v=0 = 15″, ⊙ v=+β = 12″
exchange	—	timing/Fresnel-drag derivation: without drag `nβ = 27″`; drag `β(n−1/n)`; net `β/n`. **NOT IN THE DECK** — see provenance warning below	WATER "Sum of Sun frames = 15 + 12 = βn = 27″" row
lab	—	Earth/lab frame reads `arctan(β) = 20.5″`, unchanged air → water	WATER: ⊕ v=0 = ⊕ v=−β = 20.5″ (measured)

[!warning] Provenance — the timing derivation is NOT a slide All 14 pages of `2025_Ligabue_Airy_Relativity.pdf` were rendered and checked. **The deck contains no timing / Fresnel-drag slide.** That derivation came from Franco's direct **exchange** with Alan. Cite it as the exchange, never as "slide 14". The deck's slide 14 is the wave-packet Airy example (the 15″ / 12″ / "not c/1.33" slide). An earlier revision of the vault transcription filed the timing derivation as slide 14; that has been corrected.

This matters for the video: anyone can open the PDF. Citing a slide that isn't there is a free hit for the other side, and the point stands without it — slide 13 says timing reasonings are "incorrect", and he then used one in the exchange.

[!danger] The contradiction this table exists to show Franco's relativistic water prediction is **15″ in the Sun rest frame, 12″ in the Sun moving frame** (his own values, from eq 7 and slide 13). The Earth/lab frame reads **20.5″** (what Airy measured). Three problems, stated flat: 1. **15 ≠ 12** — the two Sun-frame values disagree, so the Sun frame is not internally consistent. 2. **15 + 12 = 27″ = βn** — that sum is the classical light-slower prediction, the thing the instrument was supposed to see and did NOT. 3. **Neither 15 nor 12 is 20.5** — the Sun frame never reproduces the measurement. Air crosses the same two frames with one velocity β (→ 30 km/s). Franco's water transform $\theta_2' = \theta_2 + \beta n$ needs βn (→ 40 km/s) between the SAME two frames: a boost velocity set by the liquid in the tube. Reciprocity fails. That is the aether tell.

[!success] Verified against the PDF (2026-07-13) All 14 pages rendered and read. The 15″/12″ split and the "not c/1.33" line are on **slide 14**; the timing-argument disavowal and the rest-frame retreat are on **slide 13**; `sinθ = −β/n` (eq 7) is on **slide 3**. Slide-pin corrections applied to this table and to the vault transcription.

Backlinks

- [[2025_Ligabue_Airy_Relativity]] (vault, full slide transcription)
- [[1871_Airy_Supposed_Alteration_Aberration]]

Tags

#source #franco #verify